

how much prior volume was compressed to get there.

That second piece is precisely what makes the normalization difficult.

This is why the $L(X)$ representation is so useful. By defining

$$X(\lambda) = \int_{L(\theta) > \lambda} \pi(\theta) d\theta,$$

the high-dimensional integral becomes

$$Z = \int_0^1 L(X) dX.$$

That looks like an easy one-dimensional integral. The hard part has not disappeared, however: it has been moved into determining the relationship between L and the enclosed prior volume X .

Nested sampling is ingenious because it estimates that volume statistically. With N_{live} live points, successive volumes satisfy approximately

$$X_i = t_i X_{i-1}, \quad t_i \sim \text{Beta}(N_{\text{live}}, 1),$$

so one can progressively march from

$$X \sim 1$$

into the tiny-volume, high-likelihood regions and estimate

$$Z \approx \sum_i L_i \Delta X_i.$$

This directly attacks the aspect of evidence calculation that is hardest for ordinary Monte Carlo: **accurately measuring extremely small prior volumes associated with increasingly high likelihood contours.**

A particularly useful way to summarize the issue is:

Evidence calculation is hard because it requires measuring volumes, not just

That is also why your two schematic figures are useful pedagogically: the first contrasts naive parameter-space quadrature with integration in X , while the second makes explicit that nested sampling's real job is to infer the sequence of shrinking prior volumes X_i .

📄 🗨️ ⬆️ ↺ ...

Today 7:56 AM

What are the physics analogies of nested sampling to statistical mechanics?

There is a very direct statistical-mechanics analogy, and in fact it is one of the cleanest ways to understand nested sampling.

Start from the Bayesian evidence

$$Z = \int d\theta \pi(\theta) L(\theta).$$

Define an “energy”

$$E(\theta) \equiv -\log L(\theta),$$

so that

$$L(\theta) = e^{-E(\theta)}.$$

Then

$$Z = \int d\theta \pi(\theta) e^{-E(\theta)}.$$

This has exactly the form of a **canonical partition function at $\beta = 1$** , with the prior $\pi(\theta)d\theta$ playing the role of the underlying phase-space measure.

More generally, introduce a temperature parameter,

$$Z(\beta) = \int d\theta \pi(\theta) e^{-\beta E(\theta)} = \int d\theta \pi(\theta) L(\theta)^\beta.$$

Then ordinary Bayesian inference corresponds to $\beta = 1$, while

$$\beta = 0$$

is just the prior.

Prior volume is a cumulative density of states

This is where nested sampling becomes particularly natural from a statistical-mechanics viewpoint.

Nested sampling defines

$$X(\lambda) = \int_{L(\theta) > \lambda} \pi(\theta) d\theta.$$

Since $E = -\log L$, the condition $L > \lambda$ is equivalent to

$$E < E_\lambda.$$

Therefore

$$X(E) = \int_{E(\theta) < E} \pi(\theta) d\theta$$

is the **cumulative phase-space volume below energy E** .

That is extremely close to the integrated density of states familiar from statistical mechanics,

$$\Omega(E) = \int d\Gamma \Theta(E - H(\Gamma)).$$

Thus the correspondences are

Bayesian inference	Statistical mechanics
θ	configuration / microstate
$-\log L(\theta)$	$E(\theta)$
$\pi(\theta)d\theta$	reference phase-space measure
$X(E)$	cumulative density of states
$L = e^{-E}$	Boltzmann factor at $\beta = 1$
Z	partition function
$-\log Z$	free energy at $\beta = 1$

(up to the usual factors of β and units).

The density-of-states formulation

If

$$g(E) = \frac{dX}{dE},$$

then

$$Z(\beta) = \int g(E)e^{-\beta E} dE.$$

This is precisely the usual density-of-states expression for the canonical partition function.

Nested sampling doesn't normally estimate $g(E)$ directly. Instead, it estimates the inverse relation

$$E(X)$$

or equivalently

$$L(X) = e^{-E(X)},$$

and computes

$$Z = \int_0^1 L(X) dX.$$

So one particularly good description is:

Nested sampling is a way of reconstructing the density-of-states information

What are the live points physically analogous to?

At iteration i , nested sampling has imposed the constraint

$$L > L_i,$$

which is

$$E < E_i.$$

The live points are therefore samples from the prior measure **restricted to the region below a specified energy ceiling**.

That looks much more like a **microcanonical-type construction** than a canonical one. It is not exactly the conventional thin-shell microcanonical ensemble $E < E_i + \delta E$, because nested sampling samples the entire interior

$$E < E_i.$$

But the analogy is strong:

nested sampling repeatedly measures the phase-space volume inside successi

In the schematic figures we have been discussing, those shrinking likelihood contours are therefore literally analogous to **constant-energy surfaces**, with the algorithm progressively moving toward lower E , or equivalently higher likelihood.

Entropy makes the analogy even stronger

Because $X(E)$ measures available volume, one can define an entropy-like quantity

$$S(E) = \log X(E).$$

More conventionally one might use

$$S_{\text{micro}}(E) = \log g(E),$$

but $\log X$ is especially natural for nested sampling.

Recall the characteristic shrinkage

$$X_i \sim e^{-i/N_{\text{live}}}.$$

Therefore

$$\log X_i \simeq -\frac{i}{N_{\text{live}}}.$$

So nested sampling is effectively stepping almost uniformly in an **entropy-like variable**,

$$\Delta \log X \simeq -\frac{1}{N_{\text{live}}},$$

rather than uniformly in energy or likelihood.

This is one reason it is so effective at moving through enormous ranges of phase-space volume.

Evidence is an energy–entropy competition

This also gives a nice statistical-mechanics interpretation of the Bayesian Occam factor.

Write schematically

$$Z = \int dE e^{S(E) - \beta E},$$

where $S(E) \sim \log g(E)$.

The dominant contribution balances

$$\underbrace{-\beta E}_{\text{favor low energy/high likelihood}}$$

against

$$\underbrace{S(E)}_{\text{favor large phase-space volume}}.$$

That is exactly the same competition that determines equilibrium in statistical mechanics.

In Bayesian language:

- high likelihood favors a small region around an excellent fit;
- prior volume favors broader regions that fit reasonably well.

Thus the Bayesian tradeoff between **fit and Occam penalty** is the analogue of the statistical-mechanical tradeoff between **energy and entropy**.

This is more than a metaphor. Mathematically they are the same structure.

Posterior and canonical ensemble

The tempered posterior is

$$p_\beta(\theta) = \frac{\pi(\theta) L(\theta)^\beta}{Z(\beta)} = \frac{\pi(\theta) e^{-\beta E(\theta)}}{Z(\beta)}.$$

This is the canonical distribution.

At

$$\begin{aligned}\beta &= 0, \\ p_0(\theta) &= \pi(\theta),\end{aligned}$$

while at

$$\beta = 1$$

you have the ordinary posterior.

As $\beta \rightarrow \infty$, the distribution concentrates on the maximum-likelihood/minimum-energy configurations.

So Bayesian tempering has the literal interpretation of **cooling a statistical-mechanical system**.

Thermodynamic integration versus nested sampling

This also clarifies the connection to another evidence method, thermodynamic integration. Since

$$Z(\beta) = \int \pi(\theta) L(\theta)^\beta d\theta,$$

we have

$$\frac{d \log Z}{d\beta} = \langle \log L \rangle_\beta = -\langle E \rangle_\beta.$$

Hence

$$\log Z(1) - \log Z(0) = \int_0^1 \langle \log L \rangle_\beta d\beta.$$

Since $Z(0) = 1$,

$$\log Z = \int_0^1 \langle \log L \rangle_\beta d\beta.$$

That is essentially a **free-energy calculation by thermodynamic integration**.

So there are two complementary physics pictures:

Thermodynamic integration:	change the temperature β
Nested sampling:	change the accessible phase-space volume

Nested sampling is thus closer in spirit to a **density-of-states / microcanonical method**, while thermodynamic integration is directly a **canonical-ensemble free-energy method**.

Phase transitions and multimodal posteriors

The analogy becomes particularly interesting for difficult Bayesian problems.

Suppose the likelihood has two separated modes. In statistical mechanics these can resemble competing phases. As β changes, the dominant contribution to

$$Z(\beta)$$

may move from one region to another.

Then quantities such as

$$\langle E \rangle_\beta$$

and

$$C(\beta) = \beta^2 (\langle E^2 \rangle_\beta - \langle E \rangle_\beta^2)$$

can show behavior analogous to thermodynamic transitions.

In nested-sampling language, abrupt changes in the structure of $L(X)$ can signal transitions between large-volume/low-likelihood and small-volume/high-likelihood regions.

So multimodality, metastability, and phase coexistence have quite literal Bayesian analogues.

The interpretation I find most useful

For the figures we have just constructed, I would explain the analogy this way:

Treat $-\log L$ as an energy. Each likelihood contour is then an energy contour. The prior volume X is the cumulative phase-space volume enclosed by that contour. Nested sampling systematically compresses this accessible phase-space volume, thereby reconstructing the density-of-states information needed to calculate the partition function—i.e. the Bayesian evidence.

And the particularly elegant chain is

$$L(\theta) \longleftrightarrow E(\theta) = -\log L(\theta) \longleftrightarrow X(E) \longleftrightarrow g(E) \longleftrightarrow Z.$$

That is much closer than a loose analogy: nested sampling can genuinely be viewed as a statistical-mechanics **density-of-states/free-energy algorithm**.



ChatGPT can make mistakes. Check important info.